# United International University

## School of Science and Engineering

Final Examination; Year 2025; Trimester: Fall
Course: BIO 3105/3107; Title: Biology for Engineers/Biology;
Sec: All
Full Marks: 40; Time: 2 hours

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*There are 5 Questions. Questions **1, 2, and 3 are mandatory**. You must also answer **either 4 or 5** (any **one**)*

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| **1.** | a. Define Brain–Computer Interface (BCI) and discuss the major components involved in a BCI system. | 4 | CO4 |
| | b. Classify T cells and mention their specific functions. | 2 | CO2 |
| | c. What is a biological system? What do nodes and edges represent in such a network? | 2 | CO3 |
| | d. Why are primary databases considered the foundation of all biological databases? Explain with one example. | 2 | CO1 |
| **2.** | a. Name two experimental methods used in structural biology and briefly explain their basic principles. | 4 | CO1 |
| | b. Illustrate and label the structure of a motor neuron and outline how an action potential is generated in a neuron. | 3 | CO2 |
| | c. Discuss whether immunity is present in a baby at birth and justify your answer. | 3 | CO2 |
| **3.** | a. Explain how vaccines help the immune system to defend against pathogens? | 4 | CO2 |
| | b. What is synthetic biology? Briefly describe the Design–Build–Test–Learn (DBTL) cycle. | 3 | CO3 |
| | c. Distinguish between two well-known databases for proteomics study. | 3 | CO1 |
| **4.** | a. What is a feedback loop in a biological system? Briefly distinguish between positive and negative feedback. | 5 | CO4 |
| | b. How do NK cells operate within the immune system, and what distinguishes them from other types of white blood cells? | 5 | CO2 |
| **5.** | a. Compare artificial neural networks and biological neural networks, highlighting their differences. | 5 | CO3 |
| | b. If you obtain an unknown DNA sequence from sequencing data, how would you identify the bacterial species and construct a phylogenetic tree? | 5 | CO4 |

CO1: Describe different biological quantities.
CO2: Apply the knowledge of biological systems in a real-life problem.
CO3: Design several biological systems with constraints.
CO4: Explain several procedures for solving biological systems within constraints.